ESGIN

WHITEPAPER & TOKNOMICS

AI-Powered Climate & Carbon Ecosystem

MISSION :
Connecting AI, ESG Data, and Carbon Markets for Real-World Impact

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Legal Notice

01. Disclaimer

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The ESGIN token described in this document is intended to function as a utility token within the ESG-IN ecosystem and is designed to support interactions within the platform. It does not represent any form of equity, ownership interest, share, security, or claim on the assets, revenue, profits, or governance rights of any company, organization, or entity unless explicitly stated otherwise.

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Participation in blockchain ecosystems and digital asset networks involves inherent risks, including technological risks, regulatory uncertainty, and market volatility. Individuals and entities are responsible for assessing these risks before participating in the ESG-IN ecosystem.

02. Risk Disclosure

Participation in blockchain-based ecosystems and digital asset networks involves various risks. These risks include, but are not limited to:

Regulatory Risk

Blockchain technologies and digital assets may be subject to evolving regulatory frameworks in different jurisdictions. Changes in laws or regulations may affect the availability or functionality of ecosystem services.

Market Risk

Digital asset markets may experience significant volatility. The value of tokens may fluctuate due to various market conditions and external factors.

Technology Risk

The ESG-IN ecosystem relies on emerging technologies including blockchain systems, artificial intelligence models, and distributed network infrastructures. These technologies may experience technical limitations, vulnerabilities, or operational disruptions.

Adoption Risk

The long-term sustainability of the ESG-IN ecosystem depends on adoption by users, ESG Bank partners, and corporate participants. If adoption is slower than expected, ecosystem growth may be affected.

Operational Risk

Environmental activity verification and ESG Bank operations depend on partnerships with local organizations and infrastructure operators. Operational challenges or infrastructure disruptions could affect ecosystem functionality.

03. Forward-Looking Statements

This whitepaper may contain forward-looking statements regarding future plans, technological development, ecosystem expansion, and environmental credit frameworks.

These statements are based on current expectations, projections, and assumptions that may change over time.

Actual outcomes may differ materially from those described in this document due to various factors including technological developments, regulatory changes, market conditions, and ecosystem adoption rates.

Forward-looking statements should not be interpreted as guarantees of future performance.

04. Artificial Intelligence Limitations

The ESG-IN ecosystem incorporates artificial intelligence systems for environmental activity verification and data analysis.

Artificial intelligence technologies may assist in identifying environmental activities such as recycling participation, waste collection, and sustainability actions based on submitted images, videos, and metadata.

However, AI systems may not guarantee absolute accuracy. Environmental activity verification may evolve as machine learning models are continuously improved and trained using additional datasets.

Users and ecosystem participants acknowledge that AI-based verification systems may have limitations and may require periodic updates to maintain system reliability.

05. Jurisdiction Notice

Access to the ESG-IN ecosystem and ESGIN token functionality may be restricted or limited in certain jurisdictions where digital asset activities are regulated or prohibited.

Individuals and entities are responsible for ensuring compliance with all applicable laws and regulations in their respective jurisdictions before participating in any ecosystem activities described in this document.

The ESG-IN ecosystem does not guarantee that its services or digital assets will be available in all jurisdictions.

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01 Executive Summary

ESG-IN is an AI-powered environmental data platform designed to convert real-world sustainability activities into verifiable ESG data.

Through the ESG-IN mobile application, users can record environmental actions such as recycling participation, waste collection, and other sustainability activities. These submissions may include images, videos, and location metadata.

Artificial intelligence analyzes this data to verify environmental activities and generate structured ESG datasets, enabling environmental participation to be recorded and validated at scale.

Traditional ESG reporting systems rely heavily on corporate disclosures or estimated sustainability metrics. As a result, many environmental activities occurring at the community level remain unmeasured and excluded from global ESG frameworks.

ESG-IN addresses this limitation by introducing an AI-driven verification infrastructure capable of transforming grassroots environmental participation into structured environmental datasets.

These datasets may support sustainability analytics, ESG reporting frameworks, environmental impact measurement, and environmental credit systems such as recycling credits and carbon credits.

01. **Key #001**
    AI-Powered ESG Data Platform
02. **Key #002**
    Global Environmental Data Infrastructure
03. **Key #003**
    ESG Data + Token Economy

Within the ecosystem, the ESG-IN platform functions as the environmental data infrastructure layer where environmental activities are recorded and verified.

The ESGIN token represents the digital asset layer of the ecosystem. While environmental activities are rewarded through an off-chain point system within the ESG-IN platform, the ESGIN token supports the broader economic layer including ecosystem participation, liquidity, and value exchange.

By combining artificial intelligence, environmental infrastructure networks, and blockchain technology, ESG-IN aims to build a scalable global environmental data infrastructure where sustainability participation becomes measurable, verifiable, and economically integrated into emerging ESG markets.

02 AI-Powered ESG Infrastructure

ESG-IN is an AI-powered infrastructure designed to convert real-world environmental activities into verifiable ESG data and digital environmental assets.

The platform integrates artificial intelligence, environmental data processing, and blockchain infrastructure to create a scalable ecosystem where sustainability participation becomes measurable and traceable.

Traditional ESG reporting systems rely heavily on internal corporate disclosures or estimated sustainability metrics. These methods often fail to capture grassroots environmental activities occurring within communities.

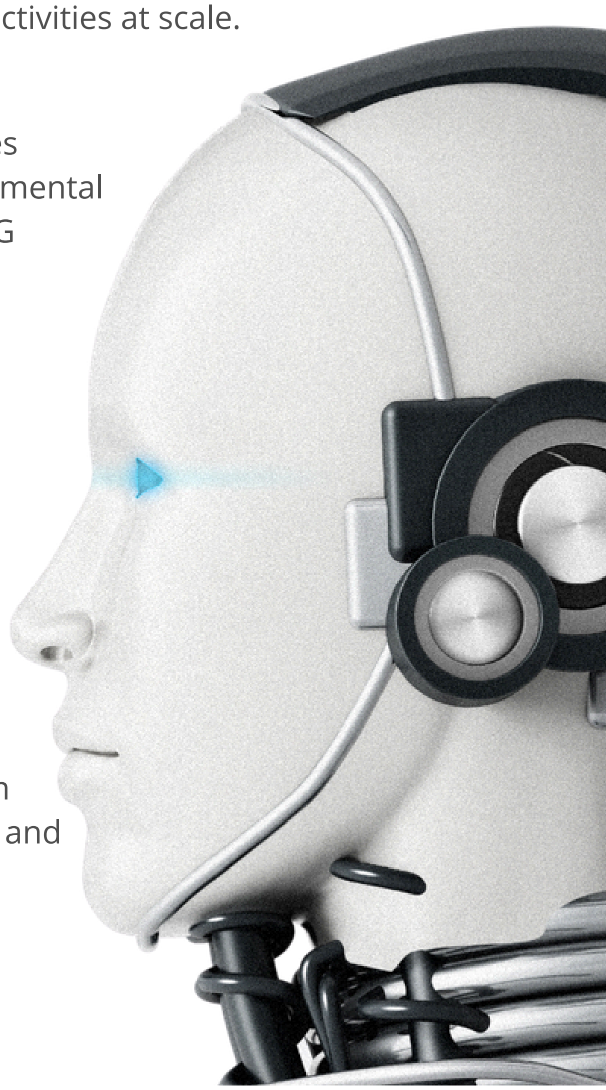
ESG-IN addresses this limitation by introducing an AI-driven verification framework capable of processing environmental activities at scale.

Through AI-based analysis of user-submitted environmental activity data, the platform generates structured ESG datasets that may support environmental credit frameworks, sustainability analytics, and ESG reporting infrastructure.

The ESG-IN ecosystem combines several core components:

- AI-based environmental activity verification
- community recycling infrastructure networks
- off-chain reward systems
- blockchain-based token infrastructure
- environmental credit aggregation

Together these elements form a unified ecosystem where environmental activity is recorded, verified, and integrated into global ESG markets.



03 Global ESG Problem Landscape

ESG Reporting Credibility Gap

Global ESG initiatives continue to expand as corporations and governments adopt sustainability commitments.

However, most ESG reporting systems rely on estimated data or internal reporting frameworks.

These systems often lack reliable mechanisms to measure environmental activity occurring at the community level.

As a result, millions of small environmental contributions remain invisible within global ESG frameworks.

Fragmented Environmental Infrastructure

Across many regions, recycling and environmental activities are performed by distributed networks of collectors, recycling facilities, and local organizations.

These systems often operate without standardized digital infrastructure. Without reliable data collection mechanisms, environmental contributions cannot be aggregated into verifiable ESG assets.

Data Limitations in Environmental Markets

Emerging environmental credit markets, including plastic credits and carbon credits, require reliable environmental datasets.

However, data collection at the community level remains inconsistent.

Without scalable verification infrastructure, environmental data cannot be transformed into standardized assets usable within ESG markets.

04 AI Verification Engine

Artificial intelligence forms the core verification mechanism within the ESG-IN ecosystem.

Users record environmental activities through the ESG-IN mobile application.

These activities may include recycling participation, waste collection, or other sustainability actions.

Users submit activity records including:

- images
- video recordings
- location metadata
- timestamp information

The AI verification engine analyzes submitted data using computer vision models trained to recognize environmental activities.

These models evaluate whether submitted content corresponds to valid sustainability actions.

The AI verification system performs several key functions:

- identification of environmental activities
- validation of activity authenticity
- classification of environmental actions
- detection of duplicate or fraudulent submissions

By automating activity verification, ESG-IN enables scalable environmental participation without requiring manual review systems.

05 Environmental Data Infrastructure

One of the primary objectives of ESG-IN is the creation of a global environmental data infrastructure.
Each verified environmental activity generates structured environmental data.

These datasets may include:

- plastic waste recovery records
- recycling participation metrics
- environmental impact indicators
- regional sustainability activity statistics

As the ecosystem expands, these datasets accumulate into a large environmental activity database.

Such datasets may support several applications:

- ESG reporting frameworks
- sustainability analytics
- environmental credit generation
- corporate ESG performance tracking

Through this infrastructure, ESG-IN transforms environmental participation into structured environmental data assets.

06 ESG Bank Network

The ESG Bank model represents the platform's integration with real-world environmental infrastructure.

Rather than replacing existing recycling networks, ESG-IN partners with local organizations operating environmental collection systems.

These partners operate under the unified ESG Bank framework.

Examples include:

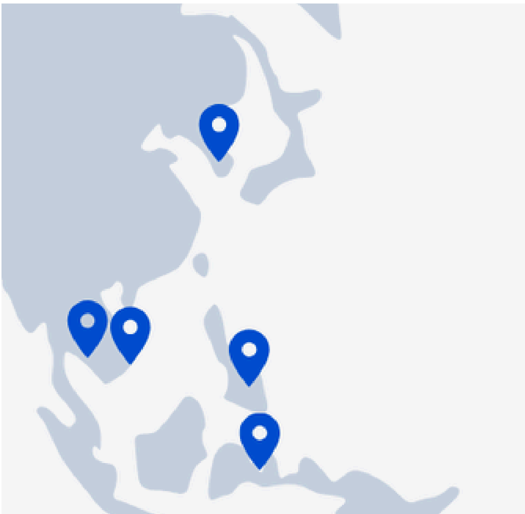
Country	Local Partner	ESG Bank Integration
Indonesia	Trash Bank	ESG Bank Indonesia
Vietnam	GRAC	ESG Bank Vietnam
Philippines	Barangay MRF	ESG Bank Philippines
Cambodia	Junkshops	ESG Bank Cambodia
South Korea	Recycling Centers	ESG Bank Korea

Notes:

This structure allows ESG-IN to scale globally while leveraging existing environmental infrastructure.

Participation Without ESG Bank Infrastructure

The ESGIN ecosystem enables environmental participation directly through the ESG-IN mobile application, even in regions where ESG Bank partners are not yet established.



Through the AI verification system, users may record and submit sustainability activities such as waste collection, recycling participation, or energy-saving actions.

This approach allows the platform to scale globally through the ESG-IN app while ESG Bank networks expand gradually as local environmental infrastructure partners.

07 Business Model

The ESG-IN ecosystem generates revenue through the integration of environmental infrastructure, ESG data services, and ecosystem participation mechanisms.

By combining AI-based environmental activity verification, community recycling infrastructure networks, and environmental data generation, ESG-IN creates multiple revenue sources across both physical recycling markets and digital ESG services.

These revenue sources support platform operations, ecosystem development, environmental programs, and long-term infrastructure expansion.

Revenue Sources

REVENUE SOURCE	DESCRIPTION	USE OF REVENUE
Material Sales	Recyclable materials aggregated through ESG Bank networks are sold to recycling processors	Supports off-chain reward points
B2B ESG Services	ESG dashboards, sustainability reporting tools, and corporate ESG campaigns	Platform operations, ecosystem development
NFT Marketplace	Transaction fees from ESG activity NFT trading	Platform development, ecosystem incentives
Environmental Credits	Environmental credits generated or acquired through verified sustainability activities	Environmental programs, ecosystem expansion
Advertising	In-app promotion of eco-friendly products and sustainability campaigns	Operations, marketing
Community Donation	Users may donate earned activity rewards, including points or ESGIN tokens, to environmental organizations and community sustainability initiatives.	Supports local environmental programs and community sustainability projects

08 ESG Bank Operator

ESG Bank operators represent local environmental infrastructure partners responsible for collecting and verifying sustainability activities within their regions.

These operators may include Trash Banks, Barangay Materials Recovery Facilities (MRFs), recycling cooperatives, and other community-based environmental networks.

Operator Rewards

Operators receive incentives for verified environmental activities processed through the ESG-IN platform.

Operator rewards are distributed through a point allocation system separate from user rewards.

Points may be converted into:

- local currency (cash settlement)
- ESGIN tokens

Operators who choose ESGIN token settlement may receive an additional 3–5% bonus incentive.

Operator Selection Criteria

To ensure long-term ecosystem alignment, ESG-IN prioritizes partnerships with operators who commit to ESGIN-based settlement mechanisms.

Only operators willing to receive rewards in ESGIN tokens may be selected as official ESG Bank partners.

This approach ensures:

- alignment with ecosystem growth
- natural token demand from operator networks
- long-term commitment from infrastructure partners

09 Two-Layer Economic System

The ESG-IN ecosystem implements a two-layer economic structure designed to separate user incentives from token supply dynamics.

Layer 1 — Off-Chain Point System

The off-chain reward layer distributes points to users for verified environmental activities.

This system provides flexibility in reward distribution while preventing uncontrolled token issuance.

Reward policies may vary across regions depending on economic conditions and ecosystem activity levels.

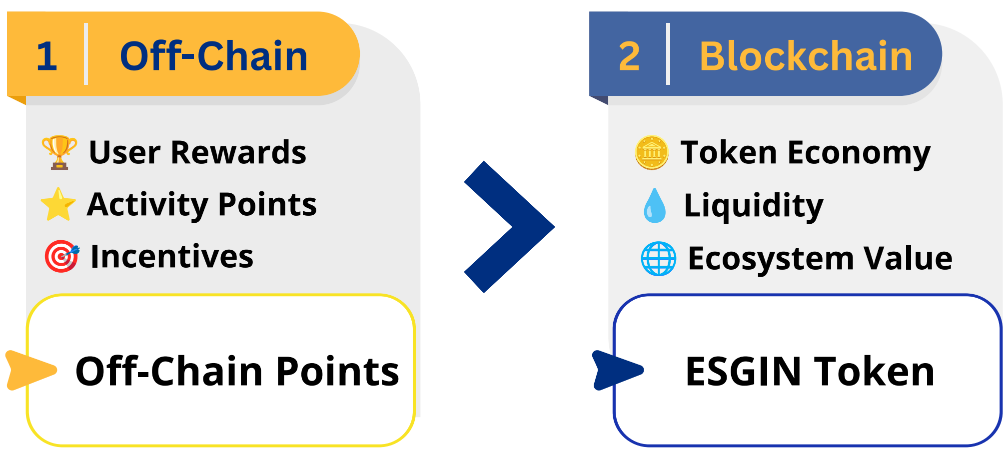
Layer 2 — ESGIN Token

The ESGIN token represents the blockchain-based asset layer of the ecosystem.

Tokens may be obtained through:

- point conversion
- ecosystem participation
- market transactions

Separating activity rewards from token issuance helps maintain balanced token circulation within the ecosystem.



10 ESG Credit Ecosystem

The ESG-IN ecosystem aims to support the development of environmental credits derived from verified sustainability activities.
The ESG-IN ecosystem aims to build a diversified set of environmental credits derived from verified sustainability activities.

♻ Plastic Recycling Credits

Plastic recycling activities recorded within the platform may contribute to plastic recycling credit frameworks.
These credits represent verified plastic waste diversion.
Supporting datasets include:

- user activity verification data
- ESG Bank collection reports
- recycling facility processing records

Carbon Credit Potential

Waste diversion activities can contribute to reductions in greenhouse gas emissions.
Environmental activity data collected through the platform may support carbon credit methodologies.
Future integrations may include environmental credit registries such as:

- Verra
- VCS
- Gold Standard

📊 Environmental Data Assets

In addition to environmental credits, ESG-IN generates valuable environmental datasets.
These datasets may support:

- sustainability reporting
- ESG analytics
- environmental impact measurement

11 Tokenomics

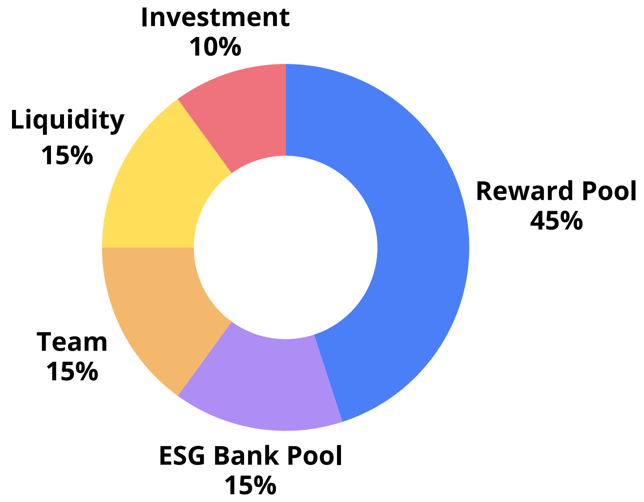
Token Overview

Token Name
ESGIN

Token Standard
ERC-20

Total Supply
1,000,000,000

Token Distribution



POOL	%	TGE	CLIFF	VESTING	PURPOSE
Reward	45%	0%	-	M1–M60	User/merchant point conversion
ESG Bank	15%	0%	-	M1–M60	ESG Bank operator rewards
Team	15%	33.33%	12M	M13–M24	Core team compensation
Liquidity	15%	66.67%	12M	M13–M24	Sales, DEX, marketing, operations
Investment	10%	100%	-	-	Strategic investments, Partnerships

TGE values indicate the percentage of each allocation created or unlocked at Token Generation Event.

Notes:

Investment: 100% of the allocation is unlocked at TGE but remains under project custody until distributed to strategic investors.
Team: 33.33% of the allocation is created at TGE and remains locked for 12 months.
Liquidity: 66.67% of the allocation is unlocked at TGE. The remaining 33.33% is released monthly from M13 to M24.

Token Release Schedule

All token allocations follow predefined vesting schedules designed to support long-term ecosystem growth and market stability. At TGE, the entire Investment allocation (10% of total supply) and 100,000,000 ESGIN (66.67% of the Liquidity allocation) are unlocked. The remaining Liquidity allocation and the unreleased Team allocation begin monthly vesting from Month 13 (M13) to Month 24 (M24). Reward Pool and ESG Bank allocations are distributed through monthly linear vesting over 60 months.

POOL	YEAR 1 (M0-M12)	YEAR 2 (M13-M24)	YEAR 3 (M25-M36)	YEAR 4 (M37-M48)	YEAR 5 (M49-M60)
Reward	9%	18%	27%	36%	45%
ESG Bank	3%	6%	9%	12%	15%
Team	0%	15%	15%	15%	15%
Liquidity	10%	15%	15%	15%	15%
Investment	10%	10%	10%	10%	10%
Max Circulating	32%	64%	76%	88%	100%

Dynamic Supply Management

ESGIN adopts a transparent token distribution model supported by predefined vesting schedules and on-chain verification. Token releases are aligned with ecosystem growth, platform adoption, and operational needs while maintaining predictable supply expansion. This approach is designed to promote long-term sustainability, support healthy market liquidity, and ensure transparency for token holders and ecosystem participants.

Circulating Supply Policy

Current circulating supply excludes all undistributed allocations held by the project, including vesting contracts, Reward Pool, ESG Bank allocation, Team allocation, locked Liquidity allocation, and Investment allocation that has not yet been distributed to investors. Although the Investment allocation is fully unlocked at TGE, it remains under project custody until distributed to investors and is therefore excluded from the current circulating supply.

12 Ecosystem Treasury Management

The ESG-IN ecosystem incorporates treasury management mechanisms designed to support long-term ecosystem sustainability and infrastructure growth.

Treasury resources are derived from ecosystem revenue streams and are allocated to support platform development, infrastructure expansion, and ecosystem incentive programs.

Treasury Allocation

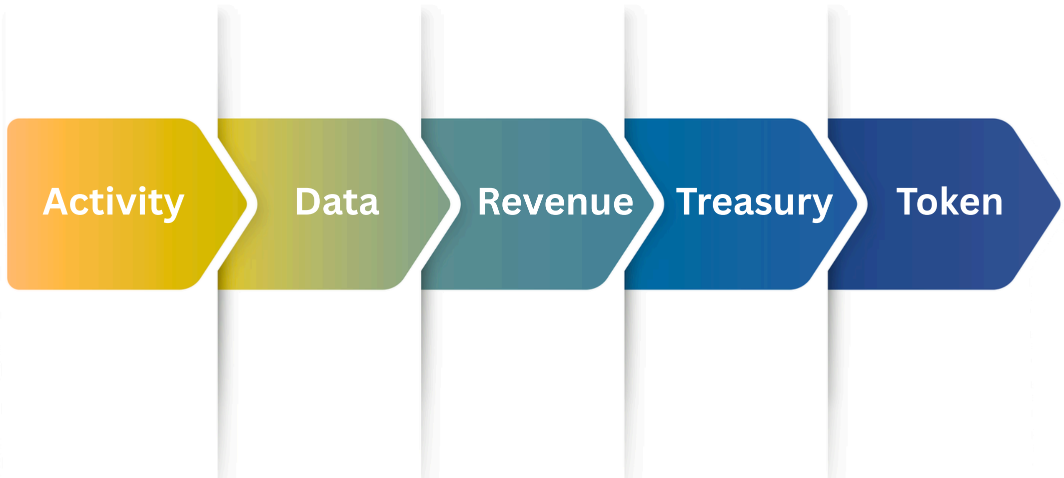
Treasury resources may be allocated across several key areas:

- platform development and technology infrastructure
- expansion of ESG Bank environmental networks
- ecosystem incentive programs
- sustainability initiatives and environmental programs
- operational support and ecosystem growth

Treasury Sustainability

The treasury mechanism is designed to support long-term ecosystem stability while enabling continued investment in environmental infrastructure and platform development.

As the ESG-IN ecosystem expands globally, treasury resources may play an important role in supporting regional growth and environmental program development.



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Governance

The ESG-IN ecosystem adopts a progressive governance model designed to balance operational stability with long-term decentralization.

During the early stages of ecosystem development, governance responsibilities are managed by the core operational team to ensure efficient platform development, infrastructure deployment, and ecosystem coordination.

As the ESG-IN ecosystem expands and community participation increases, governance mechanisms may gradually incorporate broader stakeholder participation. Over time, decentralized governance mechanisms may be introduced to allow ecosystem participants to contribute to selected decision-making processes.

This phased governance approach enables ESG-IN to maintain operational efficiency during early development while supporting the long-term goal of community participation.

Early Stage

Governance managed by the core operational team.

Growth Stage

Hybrid governance incorporating community feedback and stakeholder participation.

Mature Stage

Introduction of decentralized governance mechanisms involving ESGIN token holders.

> The governance framework may evolve

Governance Structure

In the long term, ESG-IN may implement a decentralized governance framework involving ESGIN token holders and ecosystem stakeholders.

Within this structure, governance participation may include:

- ESGIN token holders participating in governance through staking mechanisms
- voting power determined by the amount and duration of staked tokens
- regional ESG Bank representatives participating in decisions related to local environmental infrastructure

This governance model aims to align incentives between ecosystem participants while enabling community involvement in the evolution of the platform.

Governance Scope

As governance mechanisms evolve, ecosystem participants may contribute to decisions affecting the development and operation of the ESG-IN ecosystem.

DECISION AREA	EXAMPLES
Incentive Policies	Adjustment of activity reward policies and point distribution parameters
Ecosystem Expansion	Approval of expansion into new countries and selection of regional ESG Bank partners
Supply Management	Decisions related to major token supply adjustments such as token burns or long-term allocations
Treasury Allocation	Strategic ecosystem investments and environmental program funding
Ecosystem Programs	Approval of large-scale sustainability initiatives and ecosystem partnerships

14 Technology Architecture

The ESG-IN ecosystem integrates multiple technological components to support scalable environmental activity verification and data generation.

The platform architecture combines mobile applications, artificial intelligence verification systems, environmental data infrastructure, and blockchain-based asset mechanisms.

These components operate together to transform real-world environmental participation into verifiable ESG data within a unified ecosystem.

Architecture Layers

The ESG-IN technological framework consists of several integrated layers:

Application Layer

The ESG-IN mobile application enables users to record environmental activities such as recycling participation and waste collection.
This layer serves as the entry point for environmental activity data within the ESG-IN ecosystem.

AI Verification Layer

Artificial intelligence analyzes submitted activity records to validate environmental actions and detect fraudulent submissions.
This automated verification process enables large-scale environmental participation without requiring manual review systems.

Environmental Data Layer

Verified activities are converted into structured environmental datasets that support ESG analytics and sustainability reporting.
The environmental data infrastructure aggregates these datasets into a scalable environmental activity database.

Blockchain Layer

The ESGIN token functions as the digital asset layer of the ecosystem, supporting token-based participation and value exchange.
This architecture separates environmental activity rewards from token supply management while maintaining interoperability between the platform and the blockchain ecosystem.

15 Roadmap

The ESG-IN ecosystem plans to expand through a phased development strategy focused on environmental infrastructure integration, AI verification systems, and global ESG data network growth.

Each phase emphasizes the expansion of ESG Bank networks, ecosystem adoption, and environmental data infrastructure.

INDONESIA MVP (H1 2026)

The initial phase focuses on launching the ESG-IN platform and establishing the first operational environmental infrastructure network.
I50 ESG Banks, app launch, Indodax listing

REGIONAL SCALING (H2 2026)

Following the MVP launch, the ecosystem expands its environmental infrastructure network and begins regional pilot integrations.
Scale to 500 ESG Banks, Vietnam pilot (GRAC integration), corporate partnerships

SOUTHEAST ASIA EXPANSION (2027)

The third phase focuses on regional ecosystem growth and the introduction of additional ecosystem services.
Philippines & Cambodia expansion, NFT marketplace, carbon credit certification

GLOBAL ESG INFRASTRUCTURE (2028+)

The long-term vision focuses on global ecosystem expansion and the evolution of governance and environmental data infrastructure.
Global expansion, full DAO governance, IoT integration

16 Conclusion

ESG-IN introduces a new model for connecting environmental participation with digital infrastructure.

By combining artificial intelligence, environmental infrastructure networks, and blockchain technology, the platform enables real-world sustainability activities to be recorded, verified, and transformed into structured ESG data. This approach addresses a major limitation of traditional ESG reporting systems, which often rely on estimated or self-reported data.

Through scalable AI verification and environmental data infrastructure, ESG-IN aims to create a global ecosystem where environmental participation becomes measurable and traceable.

Within this ecosystem, the ESG-IN platform functions as the environmental data infrastructure layer, while the ESGIN token supports the digital asset layer enabling ecosystem participation and value exchange.

As the ecosystem expands, ESG-IN aims to contribute to the development of transparent ESG data systems and sustainable environmental infrastructure.

Join the ESG-IN Movement

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